2025 International Conference on Emerging Smart Computing and Informatics (ESCI)
AISSMS Institute of Information Technology, Pune, India. Mar 5-7, 2025

Capacitated Vehicle Routing Problem using Ant Colony Optimization and PuLP Library

Kunal Umbrani
Department of Computer Engineering and Technology
Dr.Vishwanath Karad MIT World Peace University
Pune, India
kunal.umbrani@gmail.com

Pratham Choreria
Department of Computer Engineering and Technology
Dr.Vishwanath Karad MIT World Peace University
Pune, India
prathamchoreria@gmail.com

Pradnya V.Kulkarni
Department of Computer Engineering and Technology
Dr.Vishwanath Karad MIT World Peace University
Pune, India
pradnyav.kulkarni@mitwpu.edu.in

Deven Shah
Department of Computer Engineering and Technology
Dr.Vishwanath Karad MIT World Peace University
Pune, India
deven.rshaha@gmail.com

Rhea Tarneja
Department of Computer Engineering and Technology
Dr.Vishwanath Karad MIT World Peace University
Pune, India
tarnejarhea@gmail.com

2025 International Conference on Emerging Smart Computing and Informatics (ESCI) | 979-8-3315-1568-3/25/$31.00 ©2025 IEEE | DOI: 10.1109/ESCI63694.2025.10988157

***Abstract*—Routing determines the best path for resources or data to travel, essential in sectors like networks, telecommunications and transportation. Finding the most effective routes for a fleet of vehicles to provide goods or services while taking into account variables like vehicle capacity and speed is the goal of the Vehicle Routing Problem (VRP).Solving VRP optimizes resource use, reduces costs, and boosts operational efficiency. The VRP impacts delivery schedules, vehicle maintenance and driver workloads. Optimal VRP solutions enhance delivery reliability and timeliness, crucial for customer satisfaction and competitive advantage. The Capacitated Vehicle Routing Problem (CVRP) addresses real-world challenges by ensuring efficient load balancing and preventing overloading or underutilization. CVRP solutions improve fuel economy, reduce carbon emissions and create a more cost-effective, sustainable transportation system. In order to find the best route for a fleet of capacitated vehicles, we compare the PuLP library with Ant Colony Optimization (ACO) in this study. Our study aims to evaluate the efficiency of these two methodologies in solving the capacitated vehicle routing problem (CVRP), focusing on their performance in identifying the most efficient routes under given constraints. Based on the findings, the methodology that demonstrates superior performance will be implemented in the development of a user-facing system to optimize vehicle routing for practical applications.**

Keywords— CVRP, PuLP, ACO, routing, cost, transportation

I. INTRODUCTION

Due to its high degree of complexity and practical relevance,CVRP is among the most extensively studied topics in combinatorial optimization.[1] The problem includes determining the best possible path a vehicle can take while minimizing a particular parameter. For Example, in most practical implementations, the distance that is covered by the automobile is the parameter that needs to be minimized. In traditional VRP, there are not many variables to consider, however in Capacitated Vehicle Routing, the capacity parameter, changes the approach of solving the problem. [2] Usually, the 2 main factors affecting CVRP are the vehicle capacity and the locations of the customers [3]. Because it directly affects the ultimate cost, maintaining time and cost efficiency is a top priority for all organizations. The main objective of this study is to focus on the capacitated vehicle routing problem (CVRP), define the problem appropriately and discuss the current approaches available [5].

II. LITERATURE REVIEW

In [4] the authors concentrated on streamlining the factory-in-box supply chain, especially with regard to route planning and vehicle selection. This paper offers a multi-objective optimization model for the VRP inside a factory-in-a-box to reduce operational and shipping costs. The work introduces a Hybrid Multi-Objective Evolutionary Algorithm (HMOEA) to address the multi-objective nature of MOFIBR, which produces a collection of solutions which are not controlled known as the Pareto Front. The HMOEA demonstrates its ability to tackle complex optimization issues by combining evolutionary approaches for route generation with a precise optimization strategy, thereby addressing the challenge of tackling large-scale instances of MOFIBR.

In [7], for the purpose of overcoming the issue of mobile agent routing, the authors proposed an enhanced ACO method (IACO). The ants work together using pheromone trails, an indirect communication startegy and they balance exploring new routes with using information acquired previously to find the best results. According to the experimental findings, IACO converges noticeably more quickly than the traditional ACO, simulated annealing (SA), and genetic algorithms (GA). It was discovered that the suggested algorithm produces adequate results.

The authors of [9] conducted a study on vehicle routing problems for perishable goods (VRPfPG). They identified two different subsets of issues here, which were ‘Single objective VRPfPG’ and ‘Multi-objective VRPfPG’. Considering that this problem is an NP-hard problem, researchers have tried to solve it using heuristics [10]. Techniques such as genetic algorithms and backtracking were developed to address this issue. The metaheuristic algorithm is the most popular method for resolving VRPfPG, which is demonstrated clearly in this paper.

Authorized licensed use limited to: BMS College of Engineering. Downloaded on November 12,2026 at 06:07:57 UTC from IEEE Xplore. Restrictions apply.

In [13] the authors introduced "ACO" as a pioneering model. The literature survey within the paper navigates the landscape of optimization algorithms, highlighting the shortcomings of existing approaches and emphasizing the need for innovative paradigms. Dorigo and Di Caro comprehensively explore the foundational principles of ACO, inspired by how ants behave. The survey contextualizes ACO within the broader field of evolutionary computation, shedding light on its unique features and potential applications. By critically evaluating prior methods, the authors establish the significance of their proposed meta-heuristic.

In [16] the authors proposed an "Improved ACO Model which helps in solving TSP." The authors recognize the significance of addressing large-scale instances of the problem and identify the limitations of conventional approaches. They thoroughly explore existing methodologies for TSP, emphasizing the challenges posed by increased complexity and the necessity for enhanced algorithms. By critically evaluating prior strategies, Han et al. paved the way for their proposed Improved Ant Colony Optimization (ACO) approach. The proposed Improved ACO offers a promising avenue for tackling the intricacies of this challenging problem

III. PROJECT SCOPE

A comparison study was conducted between Ant Colony Optimization and PuLP Library to obtain the best possible routes and minimum distance traveled. The next step is to build a system to allow large-scale distributors to manage their vehicles in the most efficient way possible. The distributor can input the capacities of each of their vehicles in tonnes.The system will also take the number of vehicles used to travel the routes with their given capacities.The vehicles can have constant or variable capacitie.Optimal routes will be generated for each vehicle keeping in mind the vehicle capacities.The total distance allows the distributor to understand how much each delivery cycle will cost which would help them in developing their business strategies.

IV. METHODOLOGY

A. System Design

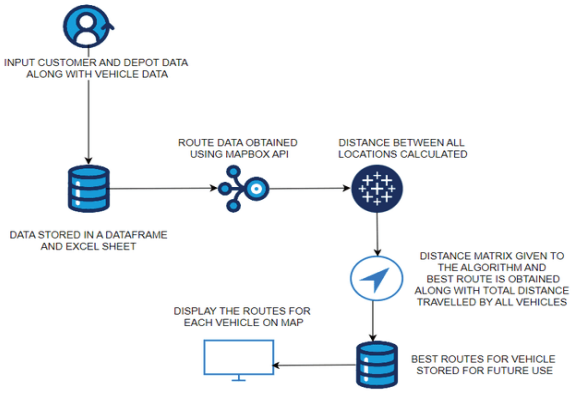


Fig. 1. System Architecture

Fig 1 shows how the system is designed. Users submit location information, vehicle specifications, and the unique requirements of each client. For documentation and additional processing, the input data is methodically saved in an Excel sheet and a DataFrame.

The system allows users to input depot and customer locations (latitude and longitude). These coordinates are added to a data frame, and a distance matrix is generated using the Mapbox API. The matrix, depicting pairwise distances between all locations, is combined with vehicle information and customer demands (also user-provided) as input for the Ant Colony Optimization model or PuLP Library algorithm to determine optimal routes.

B. Process

Using the Mapbox API, we calculate all the possible distances between all the customers and the depot. This result is stored in a 2D matrix. This information is further used in the AI model which gives us the minimum distance that needs to be traveled by a certain number of vehicles to complete the operation. Mapbox API is used in displaying the routes to the user. We used PuLP Library model and Ant Colony Optimization model for our study.A comparative analysis between both of them was performed to see which model performs better in various conditions and is best suited for our system.

```
1. Initialize parameters:
   - Set customer count to the number of customers excluding the depot
   - Define the LP problem "CVRP" as a minimization problem

2. Define decision variables for each pair of customers and each vehicle

3. Objective Function:
   - Minimize the total distance travelled by all vehicles

4. Constraints:
   a. Ensure Assignment, Depot, Flow of vehicle
   b. Ensure Capacity does not exceed the vehicle capacity
   c. Eliminate any subtours if any are getting added
5. Solve the LP problem

6. If a feasible solution is found:
   a. Print Vehicle Requirements & Total Distance
   b. Extract & Print Routes
   Otherwise:
     Print "No feasible solution found"
     Return None
```

Fig. 2. PULP Pseudo Code

Pulp - PuLP is used to model and resolve mixed-integer and linear programming issues. For specifying optimization problems and incorporating mathematical models into applications, it offers an easy-to-use interface [18]. Figure 2 describes the PuLP Pseudocode. PuLP is a flexible tool for maximizing decision-making in research and practical applications, supporting both open-source and commercial solvers [18]. It is a popular option for resolving complex issues like the CVRP because of its ease of use and versatility. PuLP is crucial for constrained optimization in a variety of academic and industrial situations [18] and seeks to find the best outcome for an objective function subject to linear constraints.

Ant Colony Optimization model – ACO model determines the most amount of vehicles required and identifies the best route for all of them[15]. ACO solves combinatorial optimization problems by simulating ant behavior, where artificial ants traverse a graph, guided by pheromone trails and heuristic data [8]. Parameters such as the number of ants, pheromone evaporation rate, and trail influence are initialized [12]. Ants construct solutions iteratively, depositing pheromones on favorable paths. Routes are saved as a list in the format: [[0, 6, 5, 4, 2, 7, 8, 0], [0, 1, 7, 2, 5, 4, 6, 9, 0]], where each sub-array represents a vehicle's route, and nodes represent locations. Routes are refined using Mapbox data and plotted to display optimal paths for each vehicle [17]. An ACO model is shown in Figure 3, in which ants build paths iteratively using pheromones and heuristic factors; pheromone levels are initialized and updated as ants explore paths, with

Authorized licensed use limited to: BMS College of Engineering. Downloaded on November 12,2026 at 06:07:57 UTC from IEEE Xplore. Restrictions apply.

evaporation applied to prevent over-concentration; the best path is identified at the end of each iteration, and its pheromone levels are reinforced to guide future ants; this process repeats over several iterations to find an optimal solution [6].

```
#Initialize parameters
num_ants,num_iterations,pheromone_decay,alpha,beta

initialize_pheromones()
for iterations in range(num_iteration):
        ant_paths=[]

        #Generate paths for each ant
        for ant in range(num_ants):
                ant_paths.append(ant_path)
                update_pheromones(ant_paths)
                decay_pheromones(pheromone_decay)

        #find the best solution
        best_path=find_best_path(ant_paths)
        update_best_path_pheromone(best_path)
function initialize_pheromones():
function generate_ant_path():
function update_pheromones(ant_paths):
function decay_pheromones(decay_rate):
function find_best_solution(ant_paths):
function update_best_path_pheromones(best_path):
```

Fig. 3. ACO Pseudo Code

Some important concepts and formulas in ACO:

1 . Initialization

- Set parameters: number of ants (m), pheromone influence (α), the degree to which the distance favorability influences the decision-making process in the algorithm(β), pheromone evaporation rate (ρ), and pheromone constant (Q).
- Initialize pheromone levels (τij) on all edges to a small positive value[13].

2 . Ants' likelihood of moving to the following node in the graph is based on the pheromone level and heuristic information.

$$P_{ij} = (\tau_{ij})^{\alpha} \cdot (\eta_{ij})^{\beta} / \Sigma_{\boldsymbol{keallowed}}(\tau_{ij})^{\alpha} \cdot (\eta_{ij})^{\beta}$$

where η_{ij} is the heuristic value [11].

3 . Pheromone Update

- Evaporate pheromone on all edges.

  $\tau_{ij} \leftarrow (1 - p) \cdot \tau_{ij}$

- Pheromone deposit is dependent on the quality of solutions.

  $\tau_{ij} \leftarrow \tau_{ij} + \sum_{k=1}^{m} \Delta\tau_{ij}$

  where $\Delta\tau_{ij}^{k} = \frac{Q}{L_k}$ if edge ij is used by ant K in its path and L_k is the distance covered by ant k [11].

4. Evaporation

- Over time, this mechanism reduces the pheromone trail intensity, preventing the convergence to less-than-ideal solutions.[11]

5. Pheromone Influence (α):

This parameter determines the significance of the pheromone trails left by previous ants [13].

6. Influence of Heuristic Information (β):

β controls the importance of heuristic information, such as the favorability of distance, when ants choose their paths [13].

7. Decay Pheromone:

The gradual decrease in pheromone trail intensity over time. This method is essential to guarantee that the algorithm effectively explores the solution space and does not prematurely converge to undesirable solutions [14].

V. Results

The study used the PuLP Library to implement the CVRP and compared its outcomes with the ACO algorithm. Tests were conducted on 10 locations in New York City, USA, and 9 in Pune, India, each with varying demands in tonnes. Additionally, 18 locations in India were analyzed for both models. Standard delivery truck capacities (7, 15, 20, 25, and 32 tonnes) were considered, with appropriate numbers of vehicles assigned to meet demands. For the ACO algorithm, parameters were fine-tuned with alpha and beta ranging from 1–3, evaporation rate at [0.5, 0.75, 1], and iterations set to [100, 150, 200].For NewYork City we found out that ACO performed slightly better than the PuLP library model in 2 out of 3 cases. For NewYork City we have a total customer demand of 52 tonnes to be fulfilled.Depot in this case is the location with coordinates (40.748817,-73.985428) with demand 0.

Latitude	Longitude	Demand(tonnes)	Name
40.748817	-73.985428	0	Empire State Building
40.743057	-73.990814	6	Flatiron Building
40.748359	-73.995250	5	Madison Square Park
40.743823	-73.989855	7	Gramercy Park
40.755161	-73.989340	5	Herald Square

Fig. 4. Locations(Latitude and longitude) for NewYork City with respective demands.

Vehicle Count	Vehicle Capacity (Tonnes)	ACO Model (Kilometers)	PuLP Library Model (Kilometers)
3	20	17.7	17.5
4	15	19.0	21.8
3	25	15.8	17.1

Fig. 5. Table showing total travel distance (in Kilometers) covered by all vehicles to fulfill the demands for NewYork City.

Authorized licensed use limited to: BMS College of Engineering. Downloaded on November 12,2026 at 06:07:57 UTC from IEEE Xplore. Restrictions apply.



Fig. 6. Routes for NewYork City for 3 vehicles with capacity of 25 tonnes using ACO model.



Fig. 7. Routes for NewYork City for 3 vehicles with capacity of 25 tonnes using PuLP model.

Latitude	Longitude	Demand	Name
18.515725	73.815313	0	MIT
18.457564	73.850735	12	PICT
18.529300	73.856825	14	COEP
18.522155	73.829947	15	Symbiosis
18.463770	73.868207	10	VIT
18.537717	73.892991	13	KP
18.564243	73.776857	9	Baner
18.538670	73.795254	8	Pashan
18.574818	73.895706	16	Airport
18.522094	73.841219	16	FC College

Fig. 8. Locations (Latitude and longitude) for Pune City in India with respective demands(Tonnes).

For New York, it is found that ACO does outperform the PuLP library in most cases. The cases where it is just edged out by PuLP can be attributed to PuLP's usefulness with small data.

For Pune City,India it was found that ACO performed better than the PuLP library model. For Pune City we have a total customer demand of 113 tonnes to be fulfilled.We have taken MIT-WPU as the depot.

Vehicle Count	Vehicle Capacity (Tonnes)	ACO Model (Kilometers)	PuLP Library Model (Kilometers)
5	25	128.2	137.4
6	20	137.7	175.4
4	32	109.5	111.3

Fig. 9. Table showing total travel distance (in Kilometers) covered by all vehicles to fulfill the demands for Pune City.

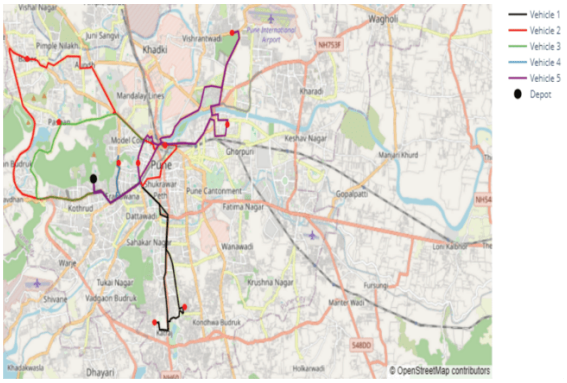


Fig. 10. Routes for Pune City for 5 vehicles with capacity of 25 tonnes using ACO model.

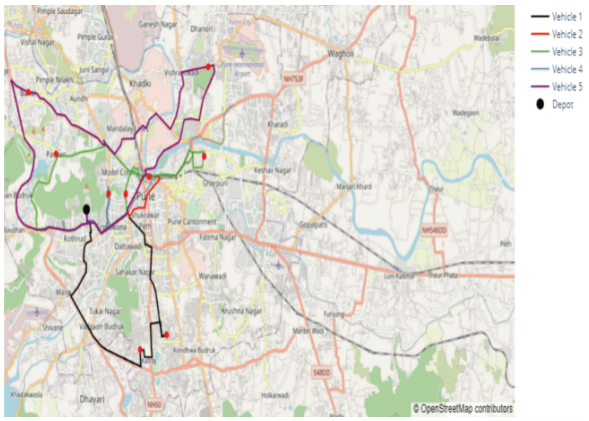


Fig. 11. Routes for Pune City for 5 vehicles with capacity of 25 tonnes using PuLP model.

In Figure 10 and 11 we can see the routes plotted for all 5 vehicles carrying 25 tonnes. Results are found for New York City as well as Pune City for vehicles with variable capacities. For Pune City the routes and distances for vehicles with variable capacities are shown below.

Vehicle Count	Vehicle Capacities (Tonnes)	ACO Model (Kilometers)	PuLP Library Model (Kilometers)
5	20,25,25,25,25	128.4	143.5
5	20,20,25,25,32	121.5	140.8
4	25,32,32,32	108.2	119.9

Fig. 12. Table showing total travel distance (in Kilometers) covered by all vehicles to fulfill the demands for Pune City.

Authorized licensed use limited to: BMS College of Engineering. Downloaded on November 12,2026 at 06:07:57 UTC from IEEE Xplore. Restrictions apply.

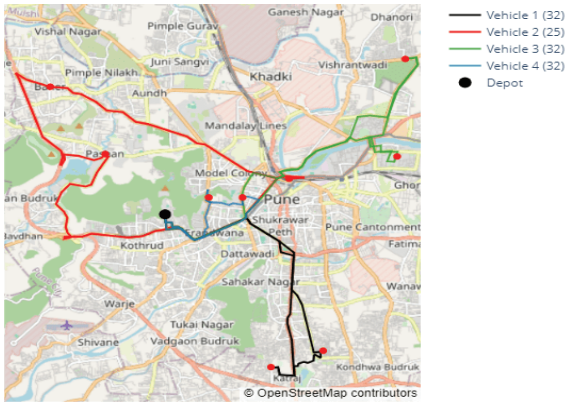


Fig. 13. Routes for Pune City for 4 vehicles with variable capacities shown on the map using PuLP model.

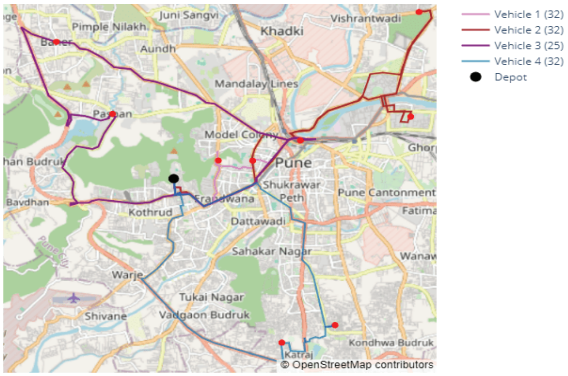


Fig. 14. Routes for Pune City for 4 vehicles with variable capacities shown on the map using ACO model.

In Figure 14 in the top right corner (legend), the color and variable capacity for respective vehicles are shown. It is observed that the ACO algorithm outperforms the PuLP library model.

It is observed that in most cases the ACO model outperforms the PuLP model for Pune City.This might be due to heuristic flexibility and adaptability of ACO model and cases where constraint handling becomes complex.We have observed that in cases where constraint complexity is less , the PuLP model gives same performance as the ACO model and even outperforms it in some cases.

Latitude	Longitude	Demand	Name
18.515725	73.815313	0	MIT
18.457564	73.850735	8	PICT
18.529300	73.856825	3	COEP
18.522155	73.829947	6	Symbiosis
18.463770	73.868207	5	VIT
18.537717	73.892991	7	KP
18.564243	73.776857	4	Baner
18.538670	73.795254	9	Pashan
18.574818	73.895706	7	Airport

Fig. 15. 18 Locations (Latitude and longitude) for Pune City in India with respective demands (Tonnes)

Vehicle Count	Vehicle Capacities (Tonnes)	ACO Model (Kilometers)
6	25,20,15,15,15,7	173.9
5	20,20,25,15,7	172.2
5	20,20,20,25,15	165.6
4	25,25,25,25	171.6

Fig. 16. Table showing total travel distance (in Kilometers) covered by all vehicles to fulfill the demands for Pune City (18 locations).

The above table shows a few combinations of vehicle count and vehicle capacities that give the minimum total distance travelled by all vehicles for Pune City with 18 locations.

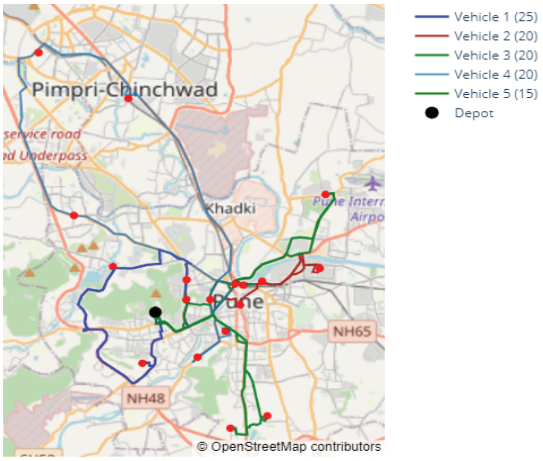


Fig. 17. Routes obtained using ACO Model for 5 vehicles with capacity of 15,20,25 tonnes.

After comprehensive analysis, the ACO model is a good method to integrate in the system.It demonstrates its efficiency compared to PuLP library model in minimizing the total distance travelled by all vehicles.

VI. LIMITATIONS

1. ***Capacity Constraints***: The primary limitation is the capacity constraint on each vehicle.A vehicle's capacity should not be exceeded by the overall demand. This constraint makes it more challenging to find optimal solutions and adds complexity to the problem.
2. ***Real-world Constraints:*** The real world introduces additional constraints, such as time windows for deliveries, service time at each customer location, and vehicle-specific constraints (e.g., maximum working hours for drivers), which may not always be considered in the basic CVRP formulation.
3. ***Solving Time:*** CVRP is an NP-hard problem, and finding optimal solutions for large instances can be computationally expensive. Practical applications often require heuristic or approximation algorithms to find ,within a reasonable time frame ,reasonably good solutions .

Authorized licensed use limited to: BMS College of Engineering. Downloaded on November 12,2026 at 06:07:57 UTC from IEEE Xplore. Restrictions apply.

VII. Future Scope

Performance Tuning and Scalability:Examine strategies for enhancing the Ant-Colony Optimization (ACO) algorithm's performance and using PuLP library. This could entail applying parallelization, adjusting settings, or making use of GPU hardware acceleration. Focus on optimizing the algorithm using PuLP library to work for a larger number of locations with less computational time and expense.

Real-time Route Adaptation:Provide a real-time adaptation system that can dynamically modify routes in response to orders or situations that change.

Hybrid Optimization Techniques:Examine the potential for integrating ACO ,PuLP algorithm model with additional optimization strategies. Hybrid methods, which take advantage of the advantages of various algorithms, can produce superior outcomes. For instance, improving the quality of the solution could involve combining the ACO or PuLP algorithm model with a local search method.

References

[1] James Fitzpatrick, Deepak Ajwani, Paula Carroll, A scalable learning approach for the capacitated vehicle routing problem, Computers & Operations Research, Volume 171,2024,106787, ISSN 0305-054 https://doi.org/10.1016/j.cor.2024.106787.

[2] Ralphs, T., Kopman, L., Pulleyblank, W. *et al.* On the capacitated vehicle routing problem. *Math. Program., Ser. B* **94**, 343–359 (2003). https://doi.org/10.1007/s10107-002-0323-0.

[3] Ibrahim, Abdullahi & Lo, Nassirou & Abdulaziz, Rabiat & Ishaya, Jeremiah. (2019). CAPACITATED VEHICLE ROUTING PROBLEM. International Journal of Research - GRANTHAALAYAH. 7. 310-327 . 10.29121/granthaalayah.v7.i3.2019.976.

[4] Junguo Cui, Lei Wu, Xiaodong Huang, Dengpan Xu, Chao Liu, Wensheng Xiao,Multi-strategy adaptable ant colony optimization algorithm and its application in robot path planning,Knowledge-Based Systems,Volume 288,2024, 111459,ISSN https://doi.org/10.1016/j.knosys.2024.111459.

[5] Zheng, J., He, K., Zhou, J., Jin, Y., & Li, C.-M. (2021). Combining Reinforcement Learning with Lin-Kernighan-Helsgaun Algorithm for the Traveling Salesman Problem. *Proceedings of the AAAI Conference on Artificial Intelligence*,*35*(14)12445-12452.https://doi.org/10.1609/aaai.v35i14.17476

[6] Jayarathna, D.G.N.D., Lanel, G.H.J. & Juman, Z.A.M.S. Industrial vehicle routing problem: a case study. *J. shipp. trd.* **7**, 6 (2022). https://doi.org/10.1186/s41072-022-00108-7.

[7] Jingan Yang, Yanbin Zhuang,An improved ant colony optimization algorithm for solving a complex combinatorial optimization problem,Applied Soft Computing,Volume 10, Issue 2,2010,Pages 653-660, ISSN 1568-4946,https://doi.org/10.1016/j.asoc.2009.08.040.

[8] Hanke Li, Xiaoming You, Sheng Liu,Multi-ant colony optimization algorithm based on finite history archiving and boxed pigs game,Applied Soft Computing,Volume 138,2023,110193,ISSN,https://doi.org/10.1016/j.asoc.2023.110193.

[9] Utama, Dana and Dewi, Shanty and Wahid, Abdul and Santoso, Imam. (2020). The vehicle routing problem for perishable goods: A systematic review. Cogent Engineering.*7*(1).https://doi.org/10.1080/23311916.2020.1816148.

[10] Chaug-Ing Hsu, Sheng-Feng Hung, Hui-Chieh Li,Vehicle routing problem with time-windows for perishable food delivery,Journal of Food Engineering,Volume 80, Issue 2,2007,Pages 465-475,ISSN 0260-8774,https://doi.org/10.1016/j.jfoodeng.2006.05.029.

[11] Y. Pei, W. Wang and S. Zhang, "Basic Ant Colony Optimization," *2012 International Conference on Computer Science and Electronics Engineering*, Hangzhou, China, 2012, pp. 665-667, doi: 10.1109/ICCSEE.2012.178.

[12] W. Deng, J. Xu and H. Zhao, "An Improved Ant Colony Optimization Algorithm Based on Hybrid Strategies for Scheduling Problem," in *IEEE Access*, vol. 7, pp. 20281-20292, 2019, doi: 10.1109/ACCESS.2019.2897580. keywords:

[13] Lei Wu, Xiaodong Huang, Junguo Cui, Chao Liu, Wensheng Xiao,Modified adaptive ant colony optimization algorithm and its application for solving path planning of mobile robot,Expert Systems with Applications,Volume 215,2023,119410,ISSN 0957-4174,https://doi.org/10.1016/j.eswa.2022.119410

[14] E. Suda and H. Iba, "Neural architecture search and weight adjustment by means of Ant Colony Optimization," *2020 5th International Conference on Intelligent Informatics and Biomedical Sciences (ICIIBMS)*, Okinawa, Japan, 2020, pp. 184-191, doi: 10.1109/ICIIBMS50712.2020.9336395.

[15] Xiaojuan Ran, Naret Suyaroj, Worawit Tepsan, Jianghong Ma, Xiangbing Zhou, Wu Deng,A hybrid genetic-fuzzy ant colony optimization algorithm for automatic K-means clustering in urban global positioning system,Engineering Applications of Artificial Intelligence,Volume 137, Part B2024,109237ISSN 0952-1976,https://doi.org/10.1016/j.engappai.2024.109237.

[16] G. K. Baydogmus, "A Parallelization Based Ant Colony Optimization for Travelling Salesman Problem," *2022 1st International Conference on Information System & Information Technology (ICISIT)*, Yogyakarta, Indonesia, 2022, pp. 166-169, doi: 10.1109/ICISIT54091.2022.9873092.

[17] M. Faisal and F. Albogamy, "Ant Colony Optimization Algorithm Enhancement for Better Performance," *2023 IEEE World AI IoT Congress (AIIoT)*, Seattle, WA, USA, 2023, pp. 0701-0710, doi: 10.1109/AIIoT58121.2023.10174442.

[18] O. Wofuru-Nyenke, “Routing and facility location optimization in a dairy products supply chain”, *futech*, vol. 3, no. 2, pp. 44–49, Oct. 2023.

Authorized licensed use limited to: BMS College of Engineering. Downloaded on November 12,2026 at 06:07:57 UTC from IEEE Xplore. Restrictions apply.